# COMPUTER NETWORKS LAB

CSE3131 · Lab Exam Quick-Write & Packet Tracer Guide

### How this PDF is organized

For each of the 10 core experiments in the supplied lab manual, the first page contains an exam-ready aim, short theory, procedure, key commands, and result. The second page gives the topology sketch and a practical step-by-step guide to building/testing it. Page numbers below refer to this PDF.

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**Note:** The experiment names and order follow the CSE3131 lab manual. The diagrams are schematic; use the exact interface names available on your Packet Tracer router/switch model. Additional experiments (Nmap, Wireshark, security and mini-project) are not included because you asked for the 10 core experiments.

# EXPERIMENT 1 · EXAM WRITE-UP

## Introduction to NetSim and Packet Tracer; networking device components

### Aim

To become familiar with Cisco Packet Tracer and NetSim and identify the main components used to design and simulate networks.

### Theory

Packet Tracer is Cisco's visual network simulator. It lets users place devices, connect them, configure interfaces and protocols, and test communication without physical hardware. NetSim is a network simulation environment for creating scenarios, running traffic, and analyzing metrics such as throughput, delay, packet delivery ratio, and packet loss. Routers forward packets between IP networks; switches forward Ethernet frames using MAC addresses; end devices generate or receive traffic.

### Procedure / Algorithm

• Open Packet Tracer and create a blank workspace.

• From the device palette, place end devices, a switch, a router, and (optionally) a server.

• Choose an appropriate cable and connect the devices. Observe interface/link indicators.

• Open a device's Config or CLI tab to inspect interfaces and settings.

• In NetSim, create a workspace, place devices, connect links, configure addressing/application traffic, run the simulation, and inspect statistics.

### Important commands

```
No mandatory CLI commands. Useful Packet Tracer checks: show ip interface brief; show running-
config; show vlan brief.
```

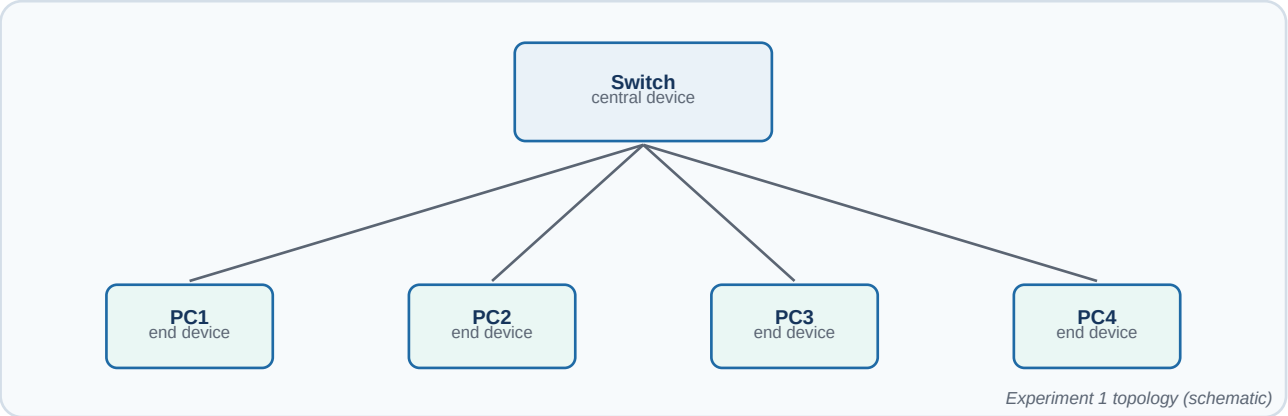
### Result / Conclusion

The simulator interface and common networking devices were identified, and a basic topology was created. In NetSim, the simulation can be run and its performance statistics inspected.

Exam tip: draw a neat labelled topology on the next page, write the IP plan, and include at least one verification command/output.

# EXPERIMENT 1 · DIAGRAM & PACKET TRACER

## Introduction to NetSim and Packet Tracer; networking device components




*Experiment 1 topology (schematic)*

## How to draw/build it

- Open a new Packet Tracer file.
- Place one switch and three PCs; optionally add a router.
- Connect each PC to the switch with Copper Straight-Through cables.
- Label devices and inspect the link indicators. This is a basic LAN demonstration.
- For a NetSim diagram, show a router, switch, end devices, links, traffic source, and performance analyzer.

## Verification checklist

Devices are placed, links are connected, and the simulation/workspace is saved.

# EXPERIMENT 2 · EXAM WRITE-UP

## Network utility commands: PING, NETSTAT, IPCONFIG, IFCONFIG, ARP, TRACERT, NSLOOKUP, PATHPING

### Aim

To use network utility commands to inspect connectivity, IP configuration, connections, address resolution, DNS, and packet routes.

### Theory

Network utilities assist diagnosis. PING uses ICMP echo messages to test reachability. NETSTAT displays connections and network statistics. IPCONFIG (Windows) and IFCONFIG (Linux/Unix; modern systems may use ip) display interface settings. ARP shows IP-to-MAC mappings. TRACERT/TRACEROUTE shows intermediate hops. NSLOOKUP queries DNS. PATHPING combines route tracing with packet-loss/latency measurements.

### Procedure / Algorithm

• Open Windows Command Prompt (or the appropriate terminal).

• Run ping 127.0.0.1 and then ping a reachable gateway or host.

• Run ipconfig /all, netstat -an, arp -a, tracert example.com, nslookup example.com, and pathping example.com as supported.

• On Linux, use ifconfig or ip addr, traceroute, and equivalent utilities.

• Record the output and explain what each command reveals. Some commands require administrator rights or an Internet connection.

### Important commands

```
Windows examples:
ping 192.168.1.1
ipconfig /all
netstat -an
arp -a
tracert example.com
nslookup example.com
pathping example.com
Linux examples: ifconfig  |  ip addr  |  traceroute example.com
```

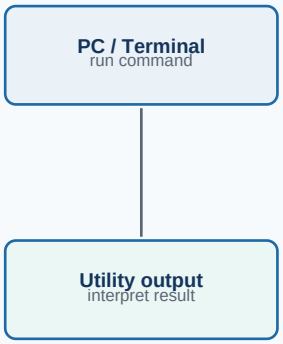
### Result / Conclusion

The commands were used to inspect connectivity, interface configuration, active connections, ARP entries, DNS resolution, and route/path performance.

Exam tip: draw a neat labelled topology on the next page, write the IP plan, and include at least one verification command/output.

# EXPERIMENT 2 · DIAGRAM & PACKET TRACER

## Network utility commands: PING, NETSTAT, IPCONFIG, IFCONFIG, ARP, TRACERT, NSLOOKUP, PATHPING




```
ping / ipconfig
arp / nslookup
```

*Experiment 2 topology (schematic)*

## How to draw/build it

• This experiment is command-line based; no Packet Tracer topology is required.

• Draw a PC connected to a switch/router and label the PC terminal as the command source.

• Write each command beside the PC and note its purpose: ping (reachability), ipconfig (address), arp (IP/MAC), nslookup (DNS), tracert/pathping (route).

• In Packet Tracer, use PC → Desktop → Command Prompt for supported commands such as ipconfig and ping.

• Do not assume Internet-dependent commands work in Packet Tracer; use a real terminal for those.

## Verification checklist

Command output is captured/interpreted; distinguish local connectivity from Internet-dependent commands.

# EXPERIMENT 3 · EXAM WRITE-UP

## Configuration of PC using Packet Tracer (star topology)

### Aim

To create a star topology, configure end-device IP addresses, and verify communication using ping.

### Theory

In a star topology, every end device connects to a central device such as a hub or switch. A hub repeats signals to its ports, while a switch forwards frames based on MAC addresses. Devices in the same IP subnet can communicate through the switch without a router, provided addressing and cabling are correct.

### Procedure / Algorithm

• Place a switch (or hub) and six PCs.

• Connect every PC to the central device using Copper Straight-Through cables.

• Assign each PC a unique IP address in the same subnet, e.g. 192.168.1.2–192.168.1.7 with mask 255.255.255.0.

• Open a PC Command Prompt and ping another PC's address.

• Optionally configure switch hostname/password and inspect show commands; save configuration if required.

### Important commands

```
Example switch setup:
enable
configure terminal
hostname S1
enable secret class
end
show running-config
copy running-config startup-config

PC test: ping 192.168.1.3
```

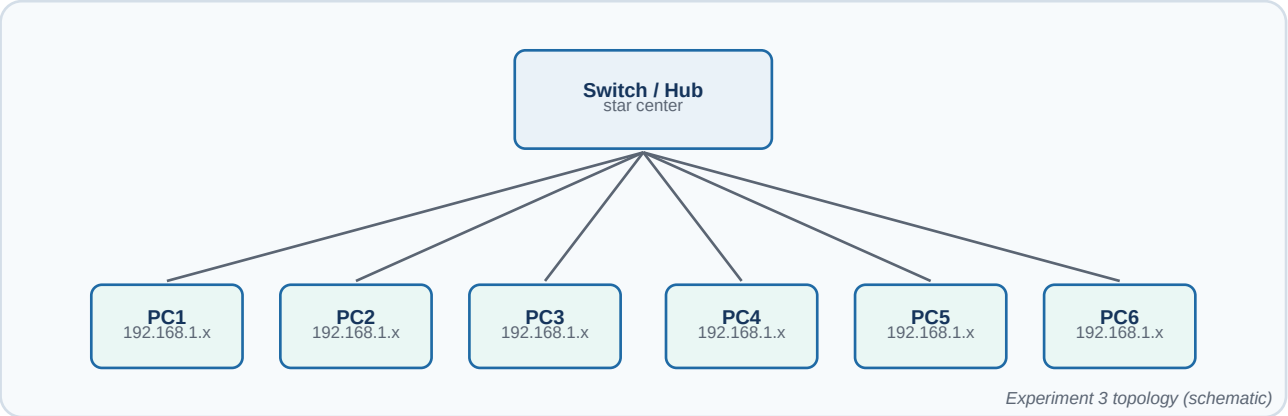
### Result / Conclusion

A star LAN was built, IP addresses were assigned, and connectivity was verified using ping. Basic switch configuration and show/copy commands were demonstrated.

Exam tip: draw a neat labelled topology on the next page, write the IP plan, and include at least one verification command/output.

# EXPERIMENT 3 · DIAGRAM & PACKET TRACER

## Configuration of PC using Packet Tracer (star topology)




*Experiment 3 topology (schematic)*

## How to draw/build it

- Place one 2960 switch at the center and six PCs around it.
- Connect each PC FastEthernet0 to a separate switch FastEthernet port.
- Set PC IPs to 192.168.1.2 through 192.168.1.7; subnet mask 255.255.255.0.
- Wait for links to turn green, then ping between two PCs.
- If configuring a switch password, remember that exact password settings depend on the lab instructions.

## Verification checklist

All links are green; PCs have unique same-subnet IPs; ping succeeds.

# EXPERIMENT 4 · EXAM WRITE-UP

## Configuration of PC using Switch and Router; DHCP

### Aim

To configure a router and switch network and use DHCP to assign IP configuration automatically to clients.

### Theory

DHCP (Dynamic Host Configuration Protocol) automatically provides clients with IP address, subnet mask, default gateway, and optionally DNS. Its four stages are DORA: Discover, Offer, Request, Acknowledgment. A router can act as a DHCP server by defining a pool and excluding reserved addresses.

### Procedure / Algorithm

• Place one router, one switch, and two or more PCs; connect them with straight-through cables.

• Configure the router LAN interface with 192.168.1.1/24 and enable it.

• Exclude addresses reserved for the gateway or static devices.

• Create a DHCP pool for 192.168.1.0/24 and set the default gateway/DNS.

• On each PC, choose Desktop → IP Configuration → DHCP.

• Verify the assigned addresses and ping the gateway/another PC.

### Important commands

```
enable
configure terminal
interface gigabitEthernet 0/0
 ip address 192.168.1.1 255.255.255.0
 no shutdown
exit
ip dhcp excluded-address 192.168.1.1 192.168.1.9
ip dhcp pool LAB
 network 192.168.1.0 255.255.255.0
 default-router 192.168.1.1
 dns-server 8.8.8.8
end
show ip dhcp binding
show ip dhcp pool
```

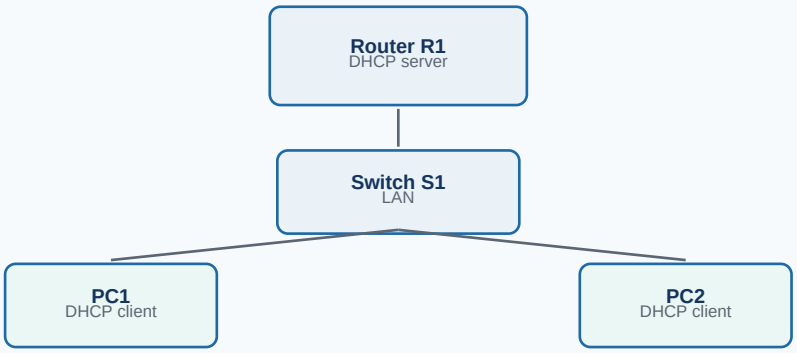
### Result / Conclusion

The router provided DHCP leases to clients, and the assigned IP settings and connectivity were verified.

Exam tip: draw a neat labelled topology on the next page, write the IP plan, and include at least one verification command/output.

# EXPERIMENT 4 · DIAGRAM & PACKET TRACER

## Configuration of PC using Switch and Router; DHCP




*Experiment 4 topology (schematic)*

## How to draw/build it

• Place one router, one 2960 switch, and two PCs.

• Connect R1 G0/0 to S1 Fa0/1; PCs to S1 Fa0/2 and Fa0/3.

• Configure the router LAN interface and DHCP pool using the commands on page 8.

• Set both PCs to DHCP in Desktop → IP Configuration.

• Check show ip dhcp binding on the router and ping 192.168.1.1 from each PC.

## Verification checklist

PCs obtain IP addresses automatically; show ip dhcp binding lists leases; gateway ping succeeds.

## Addressing reminder

Write each interface IP and subnet mask next to the link in your notebook. PCs need a default gateway when communicating beyond their own subnet. Interface names vary by router model (for example, G0/0, G0/1, or Fa0/0).

# EXPERIMENT 5 · EXAM WRITE-UP

## Router modes and basic Router/Switch commands

### Aim

To identify router command modes and practice basic router and switch configuration and verification commands.

### Theory

Cisco IOS uses hierarchical command modes. User EXEC (Router>) provides basic access. Privileged EXEC (Router#) provides diagnostic and management commands. Global configuration (Router(config)#) changes device-wide settings. Interface configuration (Router(config-if)#) configures a specific port. The switch has similar modes and commands.

### Procedure / Algorithm

• Open the device CLI and press Enter.

• Use enable to enter privileged EXEC mode.

• Use configure terminal to enter global configuration mode.

• Set hostname; select an interface; configure an IP address and use no shutdown on router interfaces.

• Use show commands to verify the configuration and copy running-config startup-config to save it.

### Important commands

```
enable
configure terminal
hostname R1
interface gigabitEthernet 0/0
 ip address 192.168.1.1 255.255.255.0
 no shutdown
exit
end
show ip interface brief
show running-config
copy running-config startup-config
```

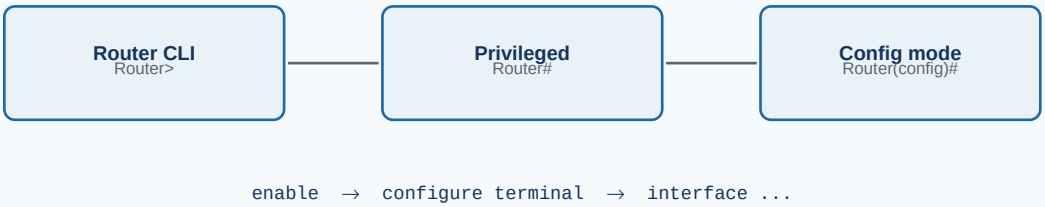
### Result / Conclusion

Router command modes were identified, an interface and hostname were configured, and show/save commands were practiced.

Exam tip: draw a neat labelled topology on the next page, write the IP plan, and include at least one verification command/output.

# EXPERIMENT 5 · DIAGRAM & PACKET TRACER

## Router modes and basic Router/Switch commands




```
enable  →  configure terminal  →  interface ...
```

*Experiment 5 topology (schematic)*

## How to draw/build it

• Place a router and a switch with one PC attached to the switch.

• Connect PC → switch → router; open the router CLI.

• Demonstrate mode transitions: enable, configure terminal, interface gigabitEthernet 0/0.

• Configure the interface address, then no shutdown.

• Run show ip interface brief and show running-config. Save with copy running-config startup-config.

## Verification checklist

CLI mode transitions are correct; interfaces are up/up; configuration is saved if required.

# EXPERIMENT 6 · EXAM WRITE-UP

## Configuration of Static Routing Protocol

### Aim

To configure static routes between two or more networks connected through routers.

### Theory

Static routing uses manually entered routes. Each route specifies the destination network and a next-hop address or exit interface. Static routes are predictable and simple for small networks but do not automatically adapt to topology changes.

### Procedure / Algorithm

• Build two LANs connected by two routers; connect the routers with an Ethernet or serial link.

• Assign IP addresses to each LAN interface and the inter-router link; enable interfaces.

• Add a route on each router to reach the remote LAN through the other router.

• Assign each PC an IP address, subnet mask, and default gateway.

• Test PC-to-PC connectivity with ping and inspect the routing table.

### Important commands

```
Example addressing: LAN-A 192.168.1.0/24; R1-R2 link 10.0.0.0/30; LAN-B 192.168.2.0/24.
R1: ip route 192.168.2.0 255.255.255.0 10.0.0.2
R2: ip route 192.168.1.0 255.255.255.0 10.0.0.1
Verify: show ip route; ping <remote-PC-IP>
```

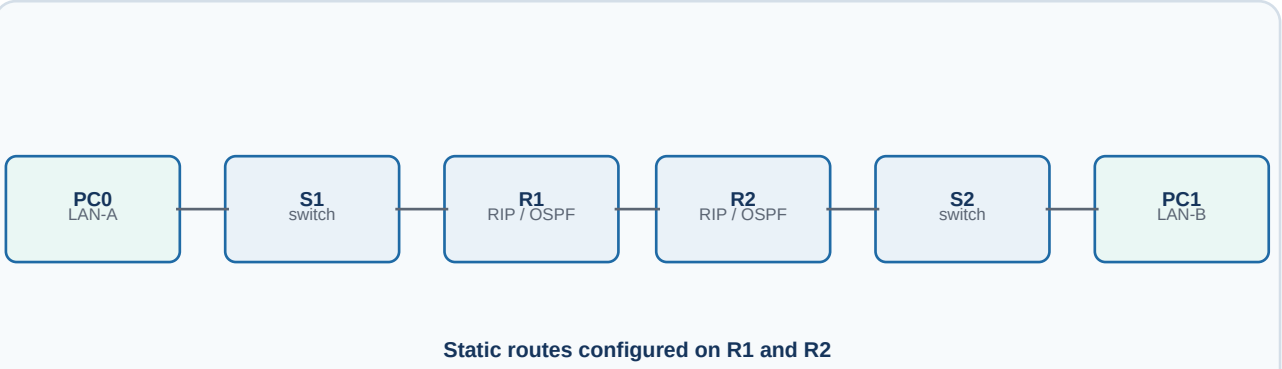
### Result / Conclusion

Static routes were configured in both directions and remote LAN connectivity was verified.

Exam tip: draw a neat labelled topology on the next page, write the IP plan, and include at least one verification command/output.

# EXPERIMENT 6 · DIAGRAM & PACKET TRACER

## Configuration of Static Routing Protocol




*Experiment 6 topology (schematic)*

## How to draw/build it

• Place two routers, two switches, and one PC on each LAN.

• Connect PC0–S1–R1 and R2–S2–PC1; connect R1 to R2.

• Use separate /24 LAN subnets and a /30 inter-router subnet; configure all interfaces.

• Add a static route on each router to the other LAN using the next-hop IP.

• Set PC gateways to their local router and test ping end to end.

## Verification checklist

Both routers have routes to the remote LAN; show ip route lists S routes; end-to-end ping succeeds.

## Addressing reminder

Write each interface IP and subnet mask next to the link in your notebook. PCs need a default gateway when communicating beyond their own subnet. Interface names vary by router model (for example, G0/0, G0/1, or Fa0/0).

# EXPERIMENT 7 · EXAM WRITE-UP

## Configuration of RIPv1 and RIPv2

### Aim

To configure RIP dynamic routing and compare RIPv1 with RIPv2 across multiple routers.

### Theory

RIP is a distance-vector routing protocol that uses hop count as its metric; the maximum usable distance is 15 hops. RIPv1 is classful and does not carry subnet masks, while RIPv2 is classless and supports VLSM/CIDR. RIP exchanges routing information periodically.

### Procedure / Algorithm

• Build a topology with at least three routers and LANs on the edge routers.

• Assign IP addresses to all LAN and inter-router interfaces; enable interfaces.

• On each router, enable RIP and advertise directly connected networks.

• For RIPv2, use version 2 and disable automatic summarization where appropriate.

• Check routing tables for routes marked R and ping between remote PCs.

### Important commands

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# no auto-summary
Router(config-router)# network 192.168.1.0
Router(config-router)# network 10.0.0.0
Repeat on each router with its directly connected networks.
Verify: show ip route; show ip protocols
```

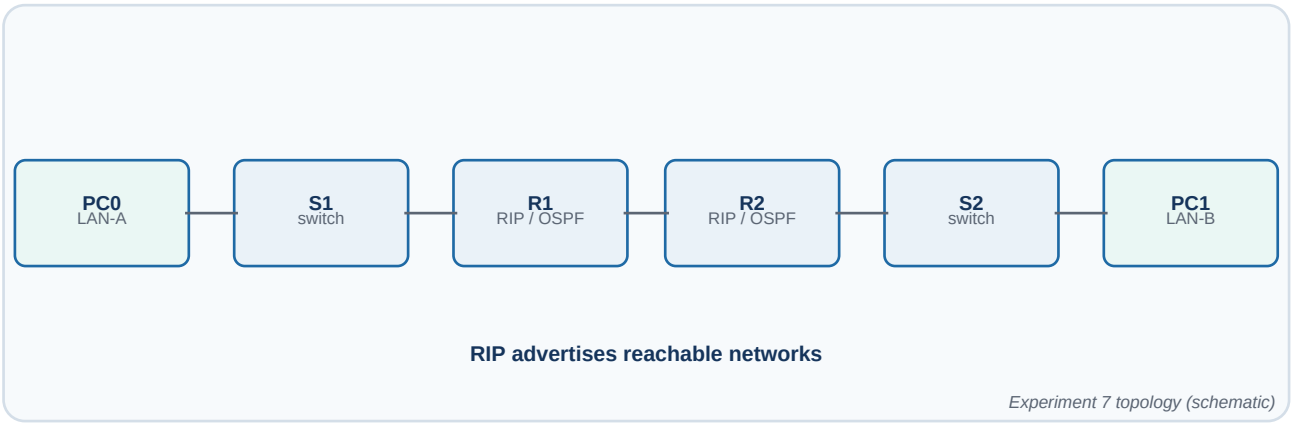
### Result / Conclusion

RIP was configured and learned routes dynamically. RIPv2 supports classless addressing and VLSM/CIDR, unlike RIPv1.

Exam tip: draw a neat labelled topology on the next page, write the IP plan, and include at least one verification command/output.

# EXPERIMENT 7 · DIAGRAM & PACKET TRACER

## Configuration of RIPv1 and RIPv2




*Experiment 7 topology (schematic)*

## How to draw/build it

• Place three routers in a line, with switches/PCs on the left and right LANs.

• Connect R1–R2 and R2–R3; configure unique link subnets and LAN subnets.

• Enable router rip on each router and advertise its connected networks.

• For RIPv2, add version 2 and no auto-summary on each router.

• Wait for updates, inspect show ip route, and ping from the left PC to the right PC.

## Verification checklist

RIP is enabled on all routers; routes appear with R; end-to-end ping succeeds.

## Addressing reminder

Write each interface IP and subnet mask next to the link in your notebook. PCs need a default gateway when communicating beyond their own subnet. Interface names vary by router model (for example, G0/0, G0/1, or Fa0/0).

# EXPERIMENT 8 · EXAM WRITE-UP

## Configuration of OSPF and troubleshooting

### Aim

To configure OSPF on multiple routers and verify dynamic route exchange and neighbor formation.

### Theory

OSPF (Open Shortest Path First) is a link-state routing protocol. Routers exchange link-state information and calculate shortest paths using cost. OSPF supports hierarchical design using areas; area 0 is the backbone. Incorrect network statements, area IDs, interface status, or IP addressing can prevent neighbor formation.

### Procedure / Algorithm

• Create a topology with two or three routers and at least two LANs.

• Configure IP addresses and bring up all interfaces.

• Enable OSPF process 1 on each router.

• Advertise each connected network with the correct wildcard mask and area, commonly area 0 for a small lab.

• Verify neighbors, routes, and end-to-end ping; troubleshoot mismatched networks/areas or down interfaces.

### Important commands

```
router ospf 1
 network 192.168.1.0 0.0.0.255 area 0
 network 10.0.0.0 0.0.0.3 area 0
Verify: show ip ospf neighbor
show ip route
show ip protocols
show ip interface brief
```

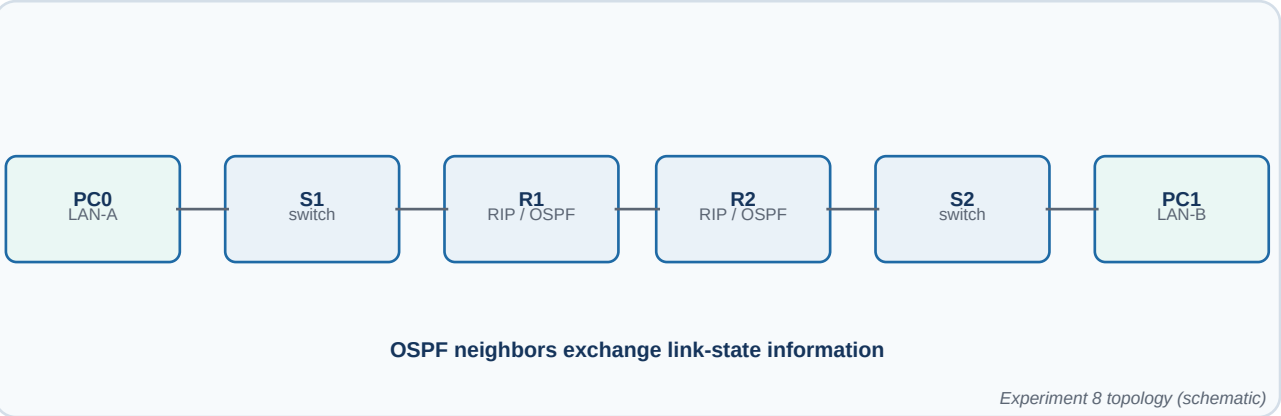
### Result / Conclusion

OSPF neighbors were established and remote routes were learned dynamically; connectivity was checked using ping.

Exam tip: draw a neat labelled topology on the next page, write the IP plan, and include at least one verification command/output.

# EXPERIMENT 8 · DIAGRAM & PACKET TRACER

## Configuration of OSPF and troubleshooting




*Experiment 8 topology (schematic)*

## How to draw/build it

• Place three routers in a line with an end LAN at each side.

• Configure all LAN and inter-router interfaces with unique IP subnets.

• On each router, enter router ospf 1 and advertise its connected networks using wildcard masks.

• Use area 0 for this simple topology.

• Check show ip ospf neighbor and show ip route. If no neighbor forms, check interfaces, IP/subnet, area, and network statements.

## Verification checklist

OSPF neighbor adjacency forms; routes appear with O; end-to-end ping succeeds.

## Addressing reminder

Write each interface IP and subnet mask next to the link in your notebook. PCs need a default gateway when communicating beyond their own subnet. Interface names vary by router model (for example, G0/0, G0/1, or Fa0/0).

# EXPERIMENT 9 · EXAM WRITE-UP

## Configuration of VLAN and troubleshooting

### Aim

To create VLANs, assign switch access ports, verify segmentation, and troubleshoot VLAN-related problems.

### Theory

A VLAN is a logical Layer-2 network that creates a separate broadcast domain on a switch. Access ports normally belong to one VLAN. Devices in different VLANs need inter-VLAN routing to communicate. VLANs improve organization and help isolate traffic.

### Procedure / Algorithm

• Place one switch and four PCs; connect each PC to a separate switch port.

• Create VLAN 10 (CSE) and VLAN 20 (Accounts).

• Assign Fa0/1–Fa0/2 to VLAN 10 and Fa0/3–Fa0/4 to VLAN 20.

• Give PCs in each VLAN IP addresses from their own subnet.

• Verify show vlan brief; test ping within a VLAN and across VLANs. Cross-VLAN ping should fail unless routing is configured.

### Important commands

```
enable
configure terminal
vlan 10
 name CSE
exit
vlan 20
 name ACCOUNTS
exit
interface range fa0/1-2
 switchport mode access
 switchport access vlan 10
exit
interface range fa0/3-4
 switchport mode access
 switchport access vlan 20
end
show vlan brief
```

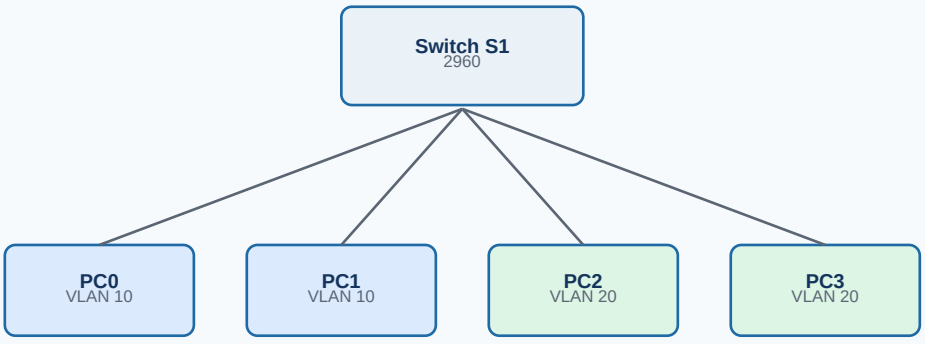
### Result / Conclusion

VLANs were created, switch ports were assigned, and logical separation was verified. Inter-VLAN communication requires a router or Laver-3 switch.

Exam tip: draw a neat labelled topology on the next page, write the IP plan, and include at least one verification command/output.

# EXPERIMENT 9 · DIAGRAM & PACKET TRACER

## Configuration of VLAN and troubleshooting




*Experiment 9 topology (schematic)*

## How to draw/build it

• Place one 2960 switch and four PCs.

• Connect PCs to Fa0/1, Fa0/2, Fa0/3, and Fa0/4.

• Configure VLAN 10 and VLAN 20 and assign the ports using the commands on page 18.

• Use 192.168.10.2/.3 for VLAN 10 and 192.168.20.2/.3 for VLAN 20; mask 255.255.255.0.

• Run show vlan brief. Same-VLAN ping should work; different-VLAN ping fails without inter-VLAN routing.

## Verification checklist

show vlan brief lists ports in correct VLANs; same-VLAN ping works; cross-VLAN ping needs routing.

## Addressing reminder

Write each interface IP and subnet mask next to the link in your notebook. PCs need a default gateway when communicating beyond their own subnet. Interface names vary by router model (for example, G0/0, G0/1, or Fa0/0).

# EXPERIMENT 10 · EXAM WRITE-UP

## NAT Protocol Configuration and troubleshooting

### Aim

To configure Network Address Translation so private LAN addresses can be translated when accessing an external network.

### Theory

NAT translates IP addresses between networks. It commonly allows private IPv4 addresses to access an external network through a public address. Static NAT maps one private address to one public address; dynamic NAT uses a pool; PAT (NAT overload) lets multiple private hosts share one public IP by using port numbers.

### Procedure / Algorithm

• Create an inside LAN connected to a router and an outside network/ISP router or server.

• Configure inside and outside interface IP addresses and enable the interfaces.

• Mark the LAN-facing interface as ip nat inside and the external-facing interface as ip nat outside.

• Create an access list matching private LAN addresses and configure NAT overload or the method specified by the instructor.

• Test from a LAN PC and inspect NAT translations/statistics.

### Important commands

```
Example PAT configuration (adjust interface names/IPs to topology):
access-list 1 permit 192.168.1.0 0.0.0.255
interface gigabitEthernet 0/0
 ip nat inside
exit
interface gigabitEthernet 0/1
 ip nat outside
exit
ip nat inside source list 1 interface gigabitEthernet 0/1 overload
Verify: show ip nat translations
show ip nat statistics
```

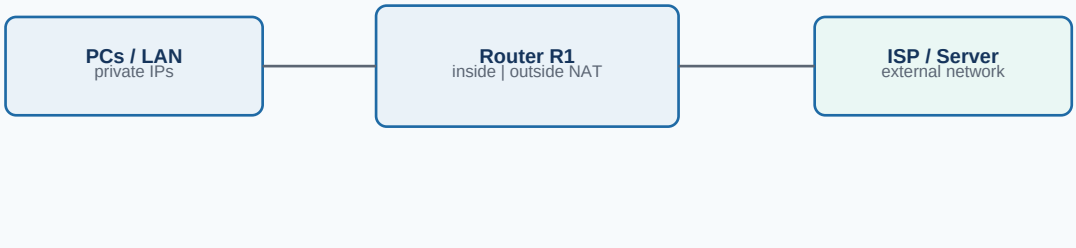
### Result / Conclusion

NAT/PAT was configured to translate inside private addresses for external communication, and translation tables/statistics were inspected.

Exam tip: draw a neat labelled topology on the next page, write the IP plan, and include at least one verification command/output.

# EXPERIMENT 10 · DIAGRAM & PACKET TRACER

## NAT Protocol Configuration and troubleshooting




*Experiment 10 topology (schematic)*

## How to draw/build it

• Place two PCs, a switch, a router, and an outside router/server.

• Connect the PCs to the switch, the switch to R1 inside interface, and R1 outside interface to the external network.

• Configure unique IP addresses and routing so the outside destination is reachable.

• Apply ip nat inside to the LAN interface, ip nat outside to the external interface, and a matching ACL/NAT rule.

• Generate traffic from a PC, then check show ip nat translations and show ip nat statistics.

## Verification checklist

Inside/outside interfaces and ACL/NAT rule are correct; NAT translations appear after traffic.

## Addressing reminder

Write each interface IP and subnet mask next to the link in your notebook. PCs need a default gateway when communicating beyond their own subnet. Interface names vary by router model (for example, G0/0, G0/1, or Fa0/0).